

NAZIRU HALILU

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SUMMARY

Agricultural engineer specializing in computer vision, geospatial machine learning, and digital agriculture. First Class Honours graduate and current Erasmus Mundus Joint Master's scholar across three European universities. Author of 4 peer reviewed publications and 14 open source research repositories archived on Zenodo.

EDUCATION

Erasmus Mundus Joint Master's Degree (EMJMD), Sustainable Agriculture 2025 to 2027

Public University of Navarre, Spain / UTAD, Portugal / Agricultural University of Athens, Greece

B.Eng., Agricultural and Bioresources Engineering 2019 to 2024

Ahmadu Bello University, Zaria, Nigeria. First Class Honours, GPA 4.77/5.00

EXPERIENCE

EMJMD Research Scholar, UTAD, Portugal Feb 2026 to Jun 2026

- Reduced input costs and improved yield prediction accuracy (R^2 0.978 to 0.995) across 65+ hectares by building GIS and machine learning pipelines for grape ripening and maize yield across three cropping systems, cutting lime demand by 2.69 tons.
- Developed a spatial kriging model (MSRK) that cut residual mapping uncertainty by 20% across 17,864 soil samples.

EMJMD Research Scholar, Public University of Navarre, Spain Sep 2025 to Jan 2026

- Built a 14 module open source pipeline for crop health screening from UAV and multispectral imagery, achieving 33.8% balanced accuracy against a 4.55% chance baseline.
- Designed a 5 stage radiometric correction framework (STADOS) that cut temporal NDVI noise by up to 39% on Sentinel 2 imagery.
- Built a Bayesian digital twin for erosion control that predicted a 47.8% reduction in annual soil loss.

Summer Intern, Computer Vision Group, Ahmadu Bello University Jun 2026 to Aug 2026

- Extended a contrastive learning framework (GACL) for crop disease recognition, reaching 64.4% classification accuracy across 19 disease classes; work submitted as first author to Biosystems Engineering.

SELECTED PUBLICATIONS

- Halilu, N., Sulaiman, J. Real Time RGB Proxy Vegetation Indexing and Texture Analysis for UAV and Handheld Crop Imagery. IJRSI, 2026.
- Halilu, N. A Hybrid CNN-BiLSTM-Attention Deep Learning Framework for Sediment Transport in an Agricultural Gully System. IJRSI, 2026.
- Halilu, N., Sulaiman, J. Sensing the Invisible: A Critical Review of NIR and Hyperspectral Analysis for Crop and Food Quality Assessment. IJRSI, 2026.

TECHNICAL SKILLS

Machine Learning:	PyTorch, TensorFlow, Keras, scikit-learn, CNNs
Geospatial / GIS:	QGIS, ArcGIS, Google Earth Engine, Sentinel imagery
Programming:	Python, MATLAB, R, Git/GitHub, Zenodo

AWARDS

Erasmus Mundus Joint Master Scholarship (2025) | Best Graduating Student, B.Eng. (2024) | Nigerian Federal Government Award for Academic Excellence (2024)